

Bashar Mqasqas

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EDUCATION

Michigan State University, College of Engineering- East Lansing, MI

GPA: 3.93

Bachelor of Science, Electrical Engineering

August 2024 – Present

- Relevant Coursework: Circuits, Electromagnetic Fields and Waves, Microprocessors, Digital Logic, Programming in C and Python, Differential Equations
- Dean's list in all eligible semesters
- Expected graduation date: May 2028

EXPERIENCE

Visteon Corporation- Van Buren Township, MI

Optics and OLED Display Intern

June 2026 – August 2026

- Configured a Keysight DAQ970A system to run a luminance degradation life test across 2 suppliers, 3 thermal conditions (room temperature, 65 °C, 80 °C), and 4 greyscale brightness levels for each display
- Prevented reversed light sensor polarity and miswired connections from causing failures across 6 2-layer measurement boards by isolating both defects on a one-off board through oscilloscope probing
- Developed a chemical-kinetics-based predictive model in Python, replicating results from standard 10,000-hour test length with ~1,500 hours of data collection
- Simulated and experimentally measured frequency response from 1 kHz to 1 MHz using TI TINA and a function generator, respectively, to design and document an IR under-panel camera (UPC) circuit schematic in TI TINA.

MSU Rocketry- East Lansing, MI

PCB Team Member

September 2024 – Present

- Engineered an accelerometer PCB using KiCad as part of an electrical system collecting flight data for student-built rockets, capturing a measured apogee of 9,964 ft in flight
- Applied a multidisciplinary approach by coordinating with payload and electronics bay teams to integrate PCB into flight systems, leading to system-level readiness for International Rocket Engineering Competition (IREC)
- Secured a 21st place finish out of 87 teams in build quality at most recent IREC

PROJECTS

Cash Register Digital Controller

- Designed a 12-bit carry-lookahead adder/subtractor as a 3-level structural Verilog hierarchy, deriving generate-propagate-carry logic from Boolean first principles to eliminate ripple-carry delay
- Implemented a 6-action Verilog FSM, plus a key-gated admin mode to control 39 fixed-protocol I/O signals, including single-level undo via secondary tally storage and two-press void confirmation
- Verified functionality through self-authored testbench spanning gate-level to full system integration, then validated final system against an instructor-authored testbench covering all required and edge-cases

Altoids Tin Headphone Amplifier

- Assembled a PCB-based op-amp audio amplifier from schematic documentation by hand soldering through-hole components, including resistors, capacitors, IC socket, and audio interfaces
- Characterized amplifier performance using a function generator (1 kHz, 200 mV peak-to-peak sine wave) and oscilloscope, confirming a 5x voltage gain at full volume with no DC offset present

USB-C Modification for DualShock 4 Controller

- Created a faster charging controller with more reliable connectivity by replacing stock micro-USB breakout board in favor of a custom USB-C counterpart
- Sourced components (resistors, connectors, USB-C port) from open-source design files, then populated the board with them using solder paste and reflow soldering
- Validated controller via multimeter and functional use, verifying proper connectivity

SKILLS

- PCB Design (ECAD) and Assembly: KiCad, reflow soldering, hand soldering
- Measurements and Instrumentation: Oscilloscope and multimeter testing
- Programming/Simulation: C, Python, MATLAB, Verilog, PSpice, TINA
- Hardware: PC assembly, hardware troubleshooting
- Autodesk (AutoCAD, Revit)
- Microsoft Office (Excel, Word, PowerPoint)